AWS WELL-ARCHITECTED FRAMEWORK

# Enterprise Ingestion & Exploitation Solution Patterns

Secure Data Ingestion and Private Tableau-to-Athena Analytic Pipelines

**Standard:** HDB Secure Architecture Guideline

**Network Scope:** Isolated HDB VPC (10.0.0.0/16)

**Data Target:** AWS S3 (SSE-KMS Secured)

# Core Architectural Assumptions

| Infrastructure Pillar | Design Assumption | Government Compliance/Standard |
|---|---|---|
| **Ingestion Isolation** | Compute engines running ETL jobs are stored exclusively in private subnets with no route to internet gateways. | IM8 Network Security Standard (Protected Segment) |
| **Compute Streaming** | Serverless ingestion tools (Glue / Lambda) ingest data.gov.sg files by streaming bytes in-chunk directly to S3. | Memory stability against Out-Of-Memory limits |
| **KMS & Auditing** | A Customer-Managed Key (CMK) controls all data read/write trails, preventing global AWS account exposure. | NIST SP 800-57 Key Management Guidelines |
| **Tableau Reachability** | Tableau Server is deployed inside a private subnet of the same VPC, or paired securely via AWS Transit Gateway. | AWS Well-Architected Security Pillar |
| **Private DNS Resolution** | VPC is configured with enableDnsSupport and enableDnsHostnames set to true. | Athena Endpoint DNS Mapping |

# Secure Ingestion Architecture Blueprint

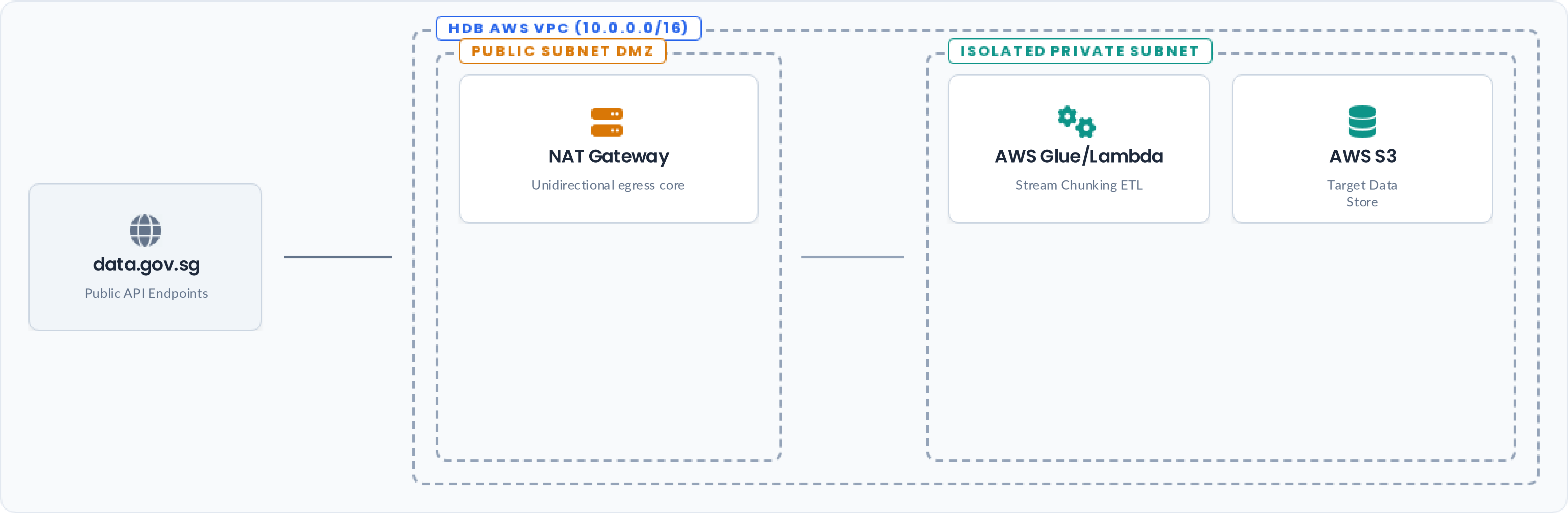


*Figure 1.0: Unidirectional network isolation mapping public API endpoints safely into highly insulated data lakes.*

# Ingestion Specifications & IAM Control

## Large File (>100MB) Streaming

Computing scripts utilize Python's requests with stream=True to split files and upload them seamlessly using S3 Multi-part Upload API, bypassing local disk limitations.

- **Zero OOM Crashes:** Payloads are processed in memory blocks of 16MB.
- **Parallel Parts:** Chunk chunks are pushed to S3 simultaneously.

## Ingestion Engine IAM Controls

The ingestion role relies strictly on the Principle of Least Privilege, isolating write capabilities to the raw landing zone bucket:

- **S3 Write Restrictions:** Restricted exclusively to s3:PutObject and multi-part upload permissions on the data target path.
- **KMS Key Cryptography:** Granted minimal keyspace permissions (kms:GenerateDataKey and kms:Encrypt) linked to a Customer Managed Key.
- **Service Trust:** Enforced through strict resource-based policies limiting execution trust to AWS Glue and Lambda.

# Secure Exploitation Blueprint

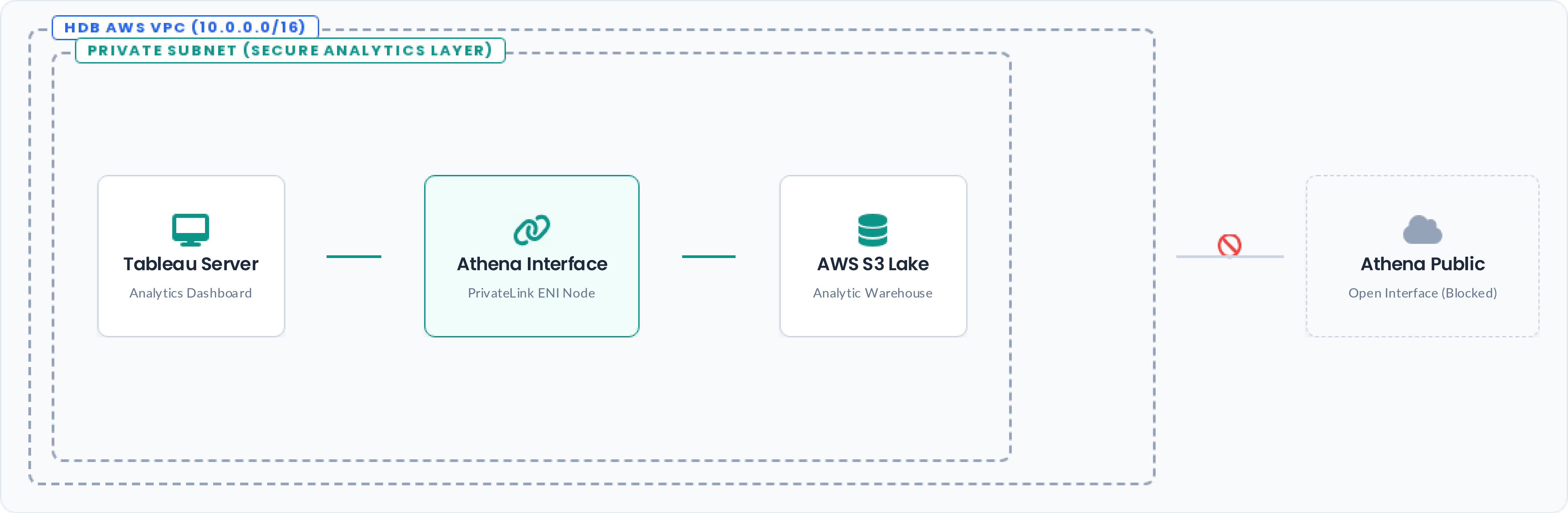


*Figure 2.0: Tableau JDBC client queries traversing AWS PrivateLink network interface cards directly, bypassing public API endpoints.*

# Exploitation Specs & IAM Access Control

## VPC Endpoint Integration

Traffic targeting the public domain is bypassed by forcing query flows through an interface-driven **Athena VPC Interface Endpoint**.

- **JDBC Connection Route:** Tableau maps endpoints inside DNS configuration parameters dynamically: vpce-1a2b3c4d-athena.ap-southeast-1.vpce.amazonaws.com

## Tableau Query IAM Access Controls

The role linked to the Tableau JDBC driver restricts interactive analytics workloads to trusted query zones only:

- **Athena Platform Controls:** Granted read execution permissions (athena:StartQueryExecution, athena:GetQueryResults) restricted strictly to the designated Tableau workgroup.
- **S3 Read Restrictions:** Limits dataset discovery actions (s3:GetObject, s3:ListBucket) solely to read paths inside the HDB warehouse store.
- **Security Audits:** Key decryption capabilities are routed dynamically through KMS configuration audits, maintaining end-to-end trace logs.